

An explicit complex polynomial retract in five variables: a consequence of affine-space cancellation failure

Alec Kriebel

ORCID: [0009-0001-9320-500X](https://orcid.org/0009-0001-9320-500X)

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Abstract

OpenAI’s explicit complex cancellation fourfold has a polynomial one-variable cylinder and is not affine four-space. We transport coefficient inclusion and evaluation at zero through its concrete cylinder coordinates. This gives a complex algebra retract of a five-variable polynomial ring, of transcendence degree four, that is not a polynomial algebra. We provide split maps and a five-component idempotent polynomial endomorphism, in factored form with no denominators, and exact symbolic certificates for the coordinate identities, including the fiber at the parameter zero. The implication from cancellation failure to a negative answer to Costa’s retract question is classical; the nonpolynomiality input and cancellation breakthrough belong to OpenAI. This note makes that consequence explicit in ambient dimension five and makes no assertion about ambient dimension four.

1 The question and the inherited input

Costa asked whether a retract of $k^{[n]}$ containing the field k must itself be a polynomial k -algebra [1]. Here a retract is specified by unital k -algebra maps $i : A \rightarrow B$ and $r : B \rightarrow A$ with $ri = 1_A$; we identify A with $i(A) \subset B$. Nagamine explicitly records the classical implication: a positive answer in $n + 1$ variables implies cancellation for A^n [2, Proposition 1.4]. His proof is coefficient inclusion followed by evaluation of the cylinder variable at zero. Nagamine settles the ambient three-variable case in characteristic zero. Earlier literature also states the polynomial-extension retract relationship [5, Introduction]. More generally, every characteristic-zero polynomial-ring retract of transcendence degree two is polynomial [3, Theorem 5.8]. The August 2026 version of Chakraborty–Pal still records the general characteristic-zero question as open for ambient $n \geq 4$ [4].

We use the following particular input, with its authorship explicit. Set

$$\begin{aligned} P &= \mathbb{C}[p, s, u, F, J], \quad x = s^2 + u^3 + p^2 F, \\ H &= x^2 F - (1 + 2sx)J - p^2 J^2 - pu, \quad A = P/(H). \end{aligned} \tag{1}$$

OpenAI [6, Theorem 1 and Sections 2–6] proves that A is an integral four-dimensional complex algebra, that $A[w] \cong \mathbb{C}^{[5]}$, and that $A \not\cong \mathbb{C}^{[4]}$. The pinned source commit is

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`

The source manuscript is dated September 23, 2026; the public repository evidence checked for this note dates from October 6, 2026. We do not equate the manuscript date with a verified public disclosure date.

The written nonpolynomiality proof is a pivotal external theorem, rather than a conclusion of the computations below. Its mechanism is worth stating precisely. The valuation degeneration is the smooth algebra

$$G = R[\tau, I/\tau^2], \quad R = \mathbb{C}[x, y, z, u]/(xy - z(z+1)),$$

where $s_R = y - x(x - u^3)$ and $I = (x - s_R^2 - u^3, z - s_R x)$. Polynomiality would give a nonzero homogeneous locally nilpotent derivation of G with a positive-degree invariant. The determinant torus bundle

$$\tilde{R} = \mathbb{C}[a, d, b, c, u]/(ac - bd - 1), \quad (x, y, z) = (ab, dc, db)$$

principalizes I to (v) , where $v = a^3b - a^2u^3 - d^2$. A normalized additive lift to its line-bundle pullback, together with the invariant, produces a nonzero locally nilpotent derivation E of $\tilde{R}$ with nonpositive shift for weights $(1, 1, -1, -1, 0)$ and $E^2(v) = 0$. The highest part for auxiliary weights $(0, 3, -3, 0, 2)$ satisfies $E'^2(d^2 + a^2u^3) = 0$. The Mason–Stothers orbit-polynomial argument and the nonsquareness of $-u^3$ in $\mathbb{C}(a, d, b)(u)$ force E' to fix a, d, u . Regularity across the determinant charts then forces its nonzero shift to be positive, a contradiction. The full proofs of the exact degeneration, bundle lift, sign extraction, and rigidity are in [6, Sections 3–6]; our dependency audit checks each of these steps.

Theorem 1. *The algebra A in (1) is a smooth integral complex algebra of transcendence degree four. There are explicit complex algebra maps $i : A \rightarrow B = \mathbb{C}[p, s, u, m, e]$ and $r : B \rightarrow A$ with $ri = 1_A$. The image $i(A)$ is not a polynomial complex algebra. The five polynomial components given in Section 3 define the idempotent algebra endomorphism ir of B and hence an idempotent endomorphism of $\mathbb{A}^5$ with image isomorphic to $\text{Spec } A$.*

Thus Costa’s assertion is false over $\mathbb{C}$ in ambient dimension five. The four-variable retract question is not settled here. The new work in this note is the explicit transport, formulas, and verification package; the conditional reduction and the cancellation input are inherited.

2 Concrete cylinder coordinates

All substitutions in this section are polynomial algebra homomorphisms. We reproduce the coordinate construction of [6, Section 2] in a form adapted to the split maps.

For independent p, s, u, F, J, w , with x as in (1), define

$$\begin{aligned} D_w &= -3u^2w + 3p^2uw^2 - p^4w^3, \\ \Phi(p) &= p, \quad \Phi(w) = w, \quad \Phi(u) = u - p^2w, \quad \Phi(s) = s + p^2xD_w, \\ \Phi(F) &= F - (1 + 2sx)D_w - p^2x^2D_w^2, \quad \Phi(J) = J - x^2D_w. \end{aligned} \tag{2}$$

The polynomial x is fixed. Replacing w by $-w$ in these substitution formulas gives the inverse. One way to verify them is to exponentiate the locally nilpotent derivation

$$\Delta(p) = 0, \quad \Delta(u) = -p^2, \quad \Delta(s) = -3p^2xu^2, \quad \Delta(F) = (6sx + 3)u^2, \quad \Delta(J) = 3x^2u^2.$$

Indeed, in the localized coordinates $y = s + x(x - u^3)$, $z = sx + p^2J$, the derivation is $-p^2\partial_u$, fixing p, x, y, z . Its displayed values are in P , so it restricts to a locally nilpotent derivation of P . Direct substitution gives $\Phi(H) = H + p^3w$. Consequently

$$\bar{\Phi} : A[w] \longrightarrow T = P[w]/(H + p^3w) \tag{3}$$

is an isomorphism, with inverse induced by Φ^{-1} .

Put $x_0 = s^2 + u^3$ and introduce

$$L = x_0^2 F - (1 + 2sx_0)J, \quad M = (1 - 2sx_0)F + 4s^2 J. \quad (4)$$

The determinant of this linear coordinate change in F, J is one; its inverse is

$$f(L) = 4s^2 L + (1 + 2sx_0)M, \quad j(L) = (2sx_0 - 1)L + x_0^2 M.$$

In the coefficient ring $K_0 = \mathbb{C}[p, s, u, M]$, set

$$\begin{aligned} Q(L) &= 2x_0 f(L)^2 - 2s f(L)j(L) - j(L)^2, \\ C &= Q(0) = M^2(2x_0 + 6sx_0^2 + 4s^2 x_0^3 - x_0^4), \\ Q(L) &= C + LQ_1(L), \quad L_* = pu - p^2 C. \end{aligned} \quad (5)$$

The factor Q_1 is a polynomial. An explicit expression, avoiding any division in the formulas, is $Q_1(L) = q_1 + q_2 L$, where

$$\begin{aligned} a_0 &= 4s^2, \quad b_0 = 1 + 2sx_0, \quad c_0 = 2sx_0 - 1, \quad d_0 = x_0^2, \\ q_1 &= M\{4x_0 a_0 b_0 - 2s(a_0 d_0 + b_0 c_0) - 2c_0 d_0\}, \\ q_2 &= 2x_0 a_0^2 - 2s a_0 c_0 - c_0^2. \end{aligned} \quad (6)$$

In these coordinates the equation is

$$H(L) = L - pu + p^2 Q(L) + p^4 f(L)^3. \quad (7)$$

Define the following polynomial in $K_0[L, w]$:

$$h = u - pQ(L) - p^3 f(L)^3 - p^2 w, \quad e_0 = -w - pf(L)^3 - Q_1(L)h. \quad (8)$$

Two exact identities are

$$H(L_*) = p^3 \{(u - pC)Q_1(L_*) + pf(L_*)^3\}, \quad (9)$$

$$p^3 e_0 - (L - L_*) = (p^2 Q_1(L) - 1)(H(L) + p^3 w). \quad (10)$$

In particular the class of e_0 in T is the unlocalized polynomial representative of $(L - L_*)/p^3$; no division is required at $p = 0$.

Define $B = \mathbb{C}[p, s, u, m, e]$, replacing M by m in expressions evaluated in B , and put

$$\begin{aligned} \ell &= pu - p^2 C + p^3 e, \quad f = f(\ell), \quad j = j(\ell), \\ W &= -e - (u - pC + p^2 e)Q_1(\ell) - pf^3. \end{aligned} \quad (11)$$

These satisfy $H(\ell) + p^3 W = 0$ identically in B . There are maps

$$q : B \longrightarrow T, \quad (p, s, u, m, e) \longmapsto (p, s, u, M, e_0),$$

and $g : T \rightarrow B$ with $(p, s, u, F, J, w) \mapsto (p, s, u, f, j, W)$.

Lemma 2. *The maps q and g are inverse isomorphisms.*

Proof. The determinant-one calculation recovers L, M from $f(L), j(L)$. Equations (11) and (10) recover L in T . Substituting $(L, w) = (\ell, W)$ into (8) gives exactly e , as a polynomial identity, and hence $gq = 1_B$ on all five generators. For the other composition, qg fixes p, s, u, M and L . The defining equation and $H(L_*) \in p^3 K_0$ then give $p^3(qg(w) - w) = 0$. The ring T is a domain and $p \neq 0$: $T \cong A[w]$, and the UFD localization argument below proves that A is a domain with nonzero p . Cancellation proves $qg(w) = w$. Thus qg fixes every generator of T . $\square$

For clarity, the domain argument does not use nonpolynomiality. Over $\mathbb{C}[p, p^{-1}]$, x, y, z, u are coordinates, with inverse

$$s = y - x(x - u^3), \quad F = p^{-2}(x - s^2 - u^3), \quad J = p^{-2}(z - sx).$$

Here $H = p^{-2}(xy - z(z + 1)) - pu$ is a coordinate, with a unit coefficient of u , so (H) is prime after localization. Since p does not divide H (its residue is the nonzero polynomial L), no factor of H can be lost by inverting p . Unique factorization proves H irreducible and prime in P . Moreover $A[p^{-1}] = \mathbb{C}[p, p^{-1}, x, y, z]$, so A has transcendence degree four.

3 Split maps and the five components

Let $j : A \rightarrow A[w]$ be coefficient inclusion and $\epsilon : A[w] \rightarrow A$ send w to zero. The actual cylinder isomorphism is $\theta = g\bar{\Phi}$. Set

$$i = \theta j, \quad r = \epsilon \bar{\Phi}^{-1} q. \quad (12)$$

The retraction is particularly simple on generators:

$$r(p) = p, \quad r(s) = s, \quad r(u) = u, \quad r(m) = M, \quad r(e) = E_A := -pF^3 - Q_1(L)(u - pQ(L) - p^3F^3) \quad (13)$$

in A , where L, M, Q, Q_1 are evaluated at the original s, u, F, J . This follows because every exponential correction vanishes at $w = 0$.

For the inclusion use (11), now in B , and set

$$\begin{aligned} X &= s^2 + u^3 + p^2 f, & D &= -3u^2 W + 3p^2 u W^2 - p^4 W^3, \\ S &= s + p^2 X D, & U &= u - p^2 W, \\ \mathcal{F} &= f - (1 + 2sX)D - p^2 X^2 D^2, & \mathcal{J} &= j - X^2 D. \end{aligned} \quad (14)$$

Then $i(p, s, u, F, J) = (p, S, U, \mathcal{F}, \mathcal{J})$. In particular $H(p, S, U, \mathcal{F}, \mathcal{J}) = 0$. For arbitrary arguments define the two polynomial functions

$$\mathcal{M}(s, u, F, J) = (1 - 2s(s^2 + u^3))F + 4s^2 J,$$

and $\mathcal{E}(p, s, u, F, J) = E_A$ from (13), using (4)–(6). The promised five components are

$$\boxed{\rho = (p, S, U, \mathcal{M}(S, U, \mathcal{F}, \mathcal{J}), \mathcal{E}(p, S, U, \mathcal{F}, \mathcal{J}))}. \quad (15)$$

This is a fully specified polynomial substitution on B , with all abbreviations defined above. It represents ir .

Proof of Theorem 1. Lemma 2 and (3) prove that θ is an actual isomorphism. Hence $ri = \epsilon\theta^{-1}\theta j = \epsilon j = 1_A$ and $(ir)^2 = i(ri)r = ir$. Its image is exactly $i(A)$, since r is surjective and i is injective.

The cited nonpolynomiality theorem [6, Theorem 1] gives $A \not\cong \mathbb{C}^{[4]}$; since A has transcendence degree four, it is not a polynomial algebra in any number of variables.

Smoothness can be checked directly. On $p \neq 0$ the displayed Laurent polynomial presentation is smooth. On $p = 0$, the partial derivatives of H with respect to F, J are x_0^2 and $-(1 + 2sx_0)$. They cannot vanish simultaneously, since $4s^2x_0^2 + (1 - 2sx_0)(1 + 2sx_0) = 1$. The hypersurface Jacobian criterion therefore proves smoothness everywhere.

On spectra, $\pi = \text{Spec}(i) : \mathbb{A}^5 \rightarrow \text{Spec } A$ and $\sigma = \text{Spec}(r) : \text{Spec } A \rightarrow \mathbb{A}^5$ obey $\pi\sigma = 1_{\text{Spec } A}$. The endomorphism with components (15) is $\sigma\pi$. Its scheme-theoretic image is the closed subscheme $\sigma(\text{Spec } A)$, with coordinate algebra $B/\ker r \cong A$. This explains both the contravariance and the image assertion. $\square$

The zero fiber is also explicit. Let $\kappa = q_1/m$, meaning the bracketed polynomial in (6), not a division operation. At $p = 0$, $W = -e - u\kappa m$ and

$$\rho(0, s, u, m, e) = (0, s, u, m + 3u^2W, -u\kappa(m + 3u^2W)).$$

Its new W is zero, so a second application fixes all components. This verifies the limiting fiber independently of localization.

4 Verification, attribution, and limits

The supplement identifies all source hashes, the exact dependency statements, audit scopes, and reproducible checks. The symbolic script uses integer polynomial arithmetic through SymPy. It checks the derivation, exponential inverse, determinant-one change, root and fifth-coordinate certificates, and the polynomial inverse identities. The other inverse composition on w uses the proved domain and p -cancellation, rather than a sample-point test. The structural proof of (12) then certifies idempotence and the image algebra. Additional exact rational examples are regression checks only; they do not prove nonpolynomiality or replace polynomial identities.

The actual upstream Lean declaration is `OAI.ComplexCancellation.main` in `OAI/Algebra/AffineCancellation/Main.lean`; its statement matches (1) and the cylinder/nonpolynomiality claim. The comparator’s `sorry` is a separate challenge statement. We inspected the actual solution sources and their semantic scope, but did not reproduce the full Lean kernel build: the local exact-version dependency build was stopped by disk limits. Accordingly this note relies on the independently audited written theorem, and claims neither a reproduced formal proof nor a formalization of the follow-on retract theorem.

Our primary-source and companion-corpus audit found no earlier explicit statement or five-component transport for this particular complex example. A failed search is not proof of priority. The general cylinder-to-retract mechanism is already explicit in Nagamine and is not claimed as new. We attribute the defining example, stabilization, and nonpolynomiality to OpenAI using its manuscript-specific citation. This is a consequence note, not an independent cancellation breakthrough.

Research disclosure. AI tools were used extensively for research, drafting, symbolic verification, and adversarial audits. The upstream work is itself AI-produced and is credited to its supplied author, OpenAI. This preprint has not undergone conventional human peer review or refereeing at publication. Automated adversarial reviews are not human peer review. No external individuals were contacted.

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